

File permissions in Linux

Project description

Through Linux, I am able to navigate and check the permissions of different directories and files, as well as change the permissions of different owner types.

Check file and directory details

Starting in the `/home/research2/` directory, I first navigated to the projects directory with the command `cd projects`. Then, I checked the permissions of the contents of the projects directory with `ls -l`. The permissions are as follows:

```
researcher2@9e3c499d53f7:~$ cd projects
researcher2@9e3c499d53f7:~/projects$ ls -l
total 20
drwx--x--- 2 researcher2 research_team 4096 Jun 18 22:05 drafts
-rw-rw-rw- 1 researcher2 research_team  46 Jun 18 22:05 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Jun 18 22:05 project_m.txt
-rw-rw-r--  1 researcher2 research_team  46 Jun 18 22:05 project_r.txt
-rw-rw-r--  1 researcher2 research_team  46 Jun 18 22:05 project_t.txt
```

Describe the permissions string

A permission string's first character indicates the file type, such as a file or directory. Characters two through four show the owner's permissions, five through seven the group's, and eight through ten the permissions for all other users. The `(r)` indicates the ability for the owner type to read the file, `(w)` to write and create new files, and `(x)` to execute files whenever applicable.

Change file permissions

In this example, I do not want the other type to have the ability to write in the `project_k.txt` file. To achieve this, I use `chmod o-w project_k.txt`. Afterwards, I can check the permissions again to see that the command worked.

```
researcher2@9e3c499d53f7:~/projects$ chmod o-w project_k.txt
researcher2@9e3c499d53f7:~/projects$ ls -l
total 20
drwx--x--- 2 researcher2 research_team 4096 Jun 18 22:05 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Jun 18 22:05 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Jun 18 22:05 project_m.txt
-rw-rw-r--  1 researcher2 research_team  46 Jun 18 22:05 project_r.txt
-rw-rw-r--  1 researcher2 research_team  46 Jun 18 22:05 project_t.txt
```

Now, the other type can only read `project_k.txt`.

Change file permissions on a hidden file

To display the permissions of hidden files, `ls -la` can be used.

```
researcher2@9e3c499d53f7:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Jun 18 22:05 .
drwxr-xr-x 3 researcher2 research_team 4096 Jun 18 22:13 ..
-rw--w---- 1 researcher2 research_team  46 Jun 18 22:05 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Jun 18 22:05 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Jun 18 22:05 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Jun 18 22:05 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 18 22:05 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 18 22:05 project_t.txt
researcher2@9e3c499d53f7:~/projects$
```

The hidden file `.project_x.txt` I do not want anyone to write on, but I still want the user and group to have read permissions. I can use the command `chmod u=r,g=r .project_x.txt` to ensure this.

```
researcher2@9e3c499d53f7:~/projects$ chmod u=r,g=r .project_x.txt
researcher2@9e3c499d53f7:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Jun 18 22:05 .
drwxr-xr-x 3 researcher2 research_team 4096 Jun 18 22:13 ..
-r--r----- 1 researcher2 research_team  46 Jun 18 22:05 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Jun 18 22:05 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Jun 18 22:05 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Jun 18 22:05 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 18 22:05 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 18 22:05 project_t.txt
```

Now, the file can only be read by the user and group owner types.

Change directory permissions

Next, I want to check the group permissions of the

`/home/researcher2/projects/drafts` directory. As shown above, those in the group have the execute permission. I will remove this with `chmod g-x drafts`.

```
researcher2@9e3c499d53f7:~/projects$ chmod g-x drafts
researcher2@9e3c499d53f7:~/projects$ ls -l
total 20
drwx----- 2 researcher2 research_team 4096 Jun 18 22:05 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Jun 18 22:05 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Jun 18 22:05 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 18 22:05 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 18 22:05 project_t.txt
```

The directory permissions have successfully been changed, and only the user can read, write, or execute files within it.

Summary

I changed multiple permissions to match the level of authorization my organization wanted for files and directories in the `projects` directory. I used `ls -la` to check the permissions, and the `chmod` command to modify the permissions on files and directories.